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SUPERSEDING
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MILITARY SPECIFICATION
PRESSURE CONTROL UNIT;
FOR M10-8 MECHANIZED FLAME THROWER

1. SCOPE

1.1 This specification covers one type of pressure control unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

DL81-1-9104	-	Pressure Control Unit.

FSC 1040

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL81-1-9104 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the pressure control unit shall conform to the specifications and drawings listed on Drawing DL81-1-9104 and subsidiary drawings.

3.2 Manufacture and assembly. The pressure control unit shall be manufactured and assembled in accordance with Drawing E81-1-9104.

3.3 Check valves leakage. The valve, Drawing B81-1-9109, shall not leak when subjected to an air pressure of 325 ± 5 pounds per square inch gage (psig), and valve, Drawing B81-1-9110, shall not leak when subjected to an air pressure of 3000 ± 50 psig for a minimum period of one minute and tested as specified in 4.4.4.1.

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3.4 Pressure regulator, manual (Drawing C81-1-9105).

3.4.1 Functioning, maximum pressure. The regulator when subjected to an inlet air pressure of 3000 ± 50 psig shall have an outlet pressure of 250 - 350 ± 5 psig for a minimum period of 5 hours when tested as specified in 4.4.4.2.

3.4.2 Functioning, minimum pressure. The regulator when subjected to an inlet air pressure of 700 ± 20 psig shall have an outlet pressure of 250 - 350 ± 5 psig for a minimum period of 5 hours when tested as specified in 4.4.4.2.

3.5 Rupture disc. The disc shall burst at a pressure between 480 psig and 540 psig when tested as specified in 4.4.4.3.

3.6 Pressure regulator, automatic (Drawing D81-1-9158). The regulator shall not leak when subjected to an air pressure of 3000 ± 50 psig at the inlet port and 350 ± 10 psig at the cover port and tested as specified in 4.4.4.4.

3.7 Pressure control unit. The unit when subjected to an inlet air pressure of 3000 ± 50 psig shall have an outlet pressure of 250 to 350 psig. The inlet pressure shall not drop more than 200 psig in a minimum period of 5 hours when tested as specified in 4.4.4.5.

3.8 Ball valve (Drawing C81-1-9101). The valve shall not leak when subjected to an air pressure of 325 ± 5 psig for a minimum period of one minute, when tested as specified in 4.4.4.6.

3.9 Valve (Drawing C81-1-9054). The valve shall not leak when subjected to an air pressure of 325 ± 5 psig for a minimum period of one minute when tested as specified in 4.4.4.7.

3.10 Preproduction. Prior to the start of regular production, a preproduction sample of pressure control units shall be produced in accordance with this specification for examination and tests (see 4.3).

3.10.1 Check valves.

3.10.1.1 Leakage, maximum pressure. The valve Drawing B81-1-9109 shall not leak when subjected to an air pressure of 550 ± 10 psig, and the valve, Drawing B81-1-9110, shall not leak when subjected to an air pressure of 4500 ± 50 psig for a minimum period of one minute when tested as specified in 4.3.3.1(a).

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3.10.1.2 Functioning. The valves shall function in accordance with table I when tested as specified in 4.3.3.1(b).

Table I. Functioning

Drawing	Temperature	Operating pressure	Flow *rate	Maximum pressure drop
B81-1-9109	minus 35° - 40°F plus 120° - 125°F	325 ± 5 psig 325 ± 5 psig	45 gpm 45 gpm	8 psig 8 psig
B81-1-9110	minus 35° - 40°F plus 120° - 125°F	3000 ± 50 psig 3000 ± 50 psig	16 gpm 16 gpm	14 psig 14 psig

*In gallons per minute (gpm).

3.10.2 Pressure regulator, manual (Drawing C81-1-9105).

3.10.2.1 Leakage, maximum pressure. The regulator shall not leak when subjected to an air pressure of 4500 ± 50 psig for a minimum period of one minute when tested as specified in 4.3.3.2(a).

3.10.2.2 Functioning. The regulator shall function in accordance with table II when tested as specified in 4.3.3.2(b).

Table II. Functioning

Temperature	Inlet pressure	Outlet pressure
Minus 45°F	3000 psig 700 psig	250 to 350 psig 250 to 350 psig
Plus 125°F	3000 psig 700 psig	250 to 350 psig 250 to 350 psig

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3.10.3 Ball valve (Drawing C81-1-9101).

3.10.3.1 Leakage, maximum pressure. The valve shall not leak when subjected to a hydrostatic pressure of 550 ± 10 psig for a minimum period of one minute, when tested as specified in 4.3.3.3(a).

3.10.3.2 Functioning. The valve shall function (turn off and on) when subjected to an air pressure of 325 ± 5 psig at a temperature of minus 35° to 40°F and at a temperature of plus 120° to 125°F when tested as specified in 4.3.3.3(b).

3.10.4 Pressure regulator, automatic (Drawing D81-1-9158).

3.10.4.1 Leakage, maximum pressure. The regulator shall not leak when subjected to an air pressure of 4500 ± 50 psig for a minimum period of one minute when tested as specified in 4.3.3.4(a).

3.10.4.2 Functioning. The regulator shall function in accordance with table III when tested as specified in 4.3.3.4(b).

Table III. Functioning

Temperature	Inlet pressure	Outlet pressure
Minus 40°F	3000 psig 700 psig	250 to 350 psig 250 to 350 psig
Plus 125°F	3000 psig 700 psig	250 to 350 psig 250 to 350 psig

3.10.5 Valve (Drawing C81-1-9054).

3.10.5.1 Leakage, maximum pressure. The valve shall not leak when subjected to a hydrostatic pressure of 550 ± 10 psig for a minimum period of one minute when tested as specified in 4.3.3.5(a).

3.10.5.2 Functioning. The valve shall function when subjected to an air pressure of 325 ± 5 psig at a temperature of minus 35° to 40°F and at a temperature of plus 120° to 125°F when tested as specified in 4.3.3.5(b).

3.10.6 Pressure control unit. The unit shall not leak when subjected to an air pressure of 4500 ± 50 psig for a minimum period of one minute when tested as specified in 4.3.3.6.

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3.11 Workmanship. The pressure control unit shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks, deformation and mutilated or kinked fittings which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of two pressure control units and ten rupture discs shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

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4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The sample pressure control units shall be examined in accordance with the list of defects (4.4.3.4) and tested as specified in 4.3.3 and 4.4.4.

4.3.2.2 For destructive test. The sample rupture discs shall be tested as specified in 4.4.4.3.

4.3.3 Tests. Components functionally tested (see 6.3) shall be cycled 400 times at the specified temperature and pressure. All leakage tests shall be conducted at ambient temperature. For air leakage tests a soap solution shall be used. For hydrostatic tests oil conforming to MIL-H-5606 shall be used. Tests shall be conducted as follows:

4.3.3.1 Check valves.

(a) Leakage.

(1) Connect the inlet end of the valve to a controlled source of air at the required pressure.

(2) Close the discharge end of the valve with a suitable fitting.

(3) Pressurize the valve with the required pressure, and observe for leakage for the specified time.

(b) Functioning.

(1) Maintain the valve at the required temperature.

(2) Connect the valve to a controlled source of hydrostatic pressure, install a flowmeter in the discharge end of valve and a suitable pressure gage.

(3) Pressurize valve and cycle as required (turn on and off to complete one cycle). Check flow rate and pressure drop.

4.3.3.2 Pressure regulator, manual.

(a) Leakage.

(1) Connect the regulator inlet to a controlled source of air at the required pressure.

(2) Close the outlet end of the regulator with a suitable fitting.

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(3) Pressurize the regulator and check for leakage for the specified time.

(b) Functioning.

(1) Maintain the regulator at the required temperature.

(2) Connect the inlet of the regulator to a controlled source of air.

(3) Connect the outlet of the regulator to a closed chamber of approximately 30 cubic inches, equipped with a pressure gage.

(4) Pressurize the regulator with the required pressure, then release the inlet and outlet pressure (one cycle), and reapply the inlet pressure and repeat as above. Check pressure gage on chamber for required outlet pressure.

4.3.3.3 Ball valve.

(a) Leakage.

(1) Connect the valve to a controlled source of hydrostatic pressure.

(2) Close the outlet end of valve with a suitable fitting.

(3) Pressurize the valve with the required pressure, and check for leakage for the specified time.

(b) Functioning.

(1) Connect the valve to a controlled source of air pressure.

(2) Maintain the valve at the required temperature.

(3) Apply the specified pressure with the valve closed. Then open the valve and vent the pressure (one cycle). Close the valve and reapply the specified pressure and repeat as above.

4.3.3.4 Pressure regulator, automatic.

(a) Leakage.

(1) Connect the regulator inlet to a controlled source of air at the required pressure.

(2) Close the outlet end of the regulator with a suitable fitting.

(3) Pressurize the regulator with the required pressure, and check for leakage for the specified time.

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(b) Functioning.

(1) Maintain the regulator at the required temperature.

(2) Connect the regulator inlet to a controlled source of air for the required pressure.

(3) Connect the outlet of the regulator to a closed chamber of approximately 30 cubic inches equipped with a pressure gage.

(4) Apply the specified pressure to the regulator. Check the pressure gage on the chamber for the required outlet pressure. Then release the inlet and outlet pressure (one cycle). For the next cycle, reapply the inlet pressure and repeat as above.

4.3.3.5 Valve (Drawing C81-1-9054).

(a) Leakage.

(1) Connect the valve to a controlled source of hydrostatic pressure.

(2) Close the outlet end of valve with a suitable plug.

(3) Pressurize the valve with the required pressure, and check for leakage for the specified time.

(b) Functioning.

(1) Connect the valve to a controlled source of air pressure.

(2) Maintain the valve at the required temperature.

(3) With the valve closed apply the specified pressure. Then open the valve, and vent the pressure (one cycle). Close the valve, reapply the specified pressure, and repeat as above.

4.3.3.6 Pressure control unit.

(a) Connect the unit to a controlled source of air for the required pressure.

(b) Close the outlet side of unit with suitable fittings.

(c) Pressurize the unit with the required pressure, and check for leakage for the specified time.

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4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. Preproduction samples shall not be used after tests. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting.

4.4.1.1 Rupture discs. A lot shall consist of the discs produced by one manufacturer, from approximately one square foot of material, taken from a sheet of material produced by one manufacturer, under essentially the same manufacturing conditions.

4.4.1.2 Valves and pressure regulators. A lot shall consist of the valves or regulators of one interfix number produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.1.3 Serial number. Each pressure control unit shall be sequentially serial-numbered.

4.4.2 Sampling.

4.4.2.1 For nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Each pressure control unit shall be examined in accordance with the list of defects (4.4.3.4). If any unit contains any defect listed in 4.4.3.4, that unit shall be rejected.

4.4.3.2 For nondestructive tests. The sample components and units shall be tested in accordance with 4.4.4 and MIL-STD-105 using an AQL of 2.5 percent defective.

4.4.3.3 For destructive test. The sample discs shall be tested in accordance with 4.4.4.3. The acceptance number is zero.

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4.4.3.4 List of defects.

(a) Pressure control unit (Drawing E81-1-9104).

(1) Component missing

(2) More than one rupture disc

(3) Pressure regulator nonfunctioning

(4) High pressure gage nonfunctioning

(5) Low pressure gage nonfunctioning

(6) Check valve nonfunctioning

(7) Fuel pressurizing valve nonfunctioning

(8) Vent valve nonfunctioning

(9) Workmanship (3.11)

(b) Preparation for delivery (section 5).

(1) Loose pack

(2) Container closure incorrect

(3) Openings into unit not sealed

4.4.4 Tests. Test shall be conducted as follows:

4.4.4.1 Check valves.

(a) Connect the outlet end of the valve to a controlled source of air at the required pressure.

(b) Pressurize valve with the required pressure.

(c) Observe for leakage for the required time (see 4.3.3).

4.4.4.2 Pressure regulator, manual.

(a) Connect the regulator inlet to a controlled source of air at the required pressure.

(b) Connect the outlet of the regulator to a suitable fitting with a pressure gage and shut-off valve.

(c) Pressurize the regulator with the required pressure and allow the regulator to stabilize for a minimum period of 5 minutes.

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(d) Close the inlet source of air valve and the outlet air valve, and check outlet pressure gage periodically for the required time.

4.4.4.3 Rupture disc. The rupture disc shall be placed in a suitable fixture and connected to a controlled source of air pressure. Gradually increase the pressure until the disc bursts, and record the pressure.

4.4.4.4 Pressure regulator, automatic.

(a) Connect the regulator inlet and cover port to a controlled source of air pressure.

(b) Plug the gage port and outlet port.

(c) Connect a short length of 1/2 inch tubing to the vent port. Place the open end of tubing under water.

(d) Slowly increase the air pressure to the required amount.

(e) Allow the regulator to stabilize for 5 minutes, and then observe the open end of the tubing for leakage, for 5 minutes ± 15 seconds.

4.4.4.5 Pressure control unit.

(a) Connect the inlet of the unit to a controlled source of air at the required pressure with a pressure gage and shut off valve in the line.

(b) Connect a pressure gage and shut off valve to the outlet of the unit with the pressure gage on the upstream side of the shut off valve. Leave the shut off valve closed.

(c) Pressurize the unit with the required pressure. Open the outlet valve sufficiently to allow air to bleed off, and allow the unit to stabilize for 5 minutes. Check the outlet pressure gage for meeting pressure requirement. Then shut off the outlet and inlet valves.

(d) Check the inlet pressure gage for a drop in pressure for the required time.

4.4.4.6 Ball valve leakage.

(a) Connect the valve to a controlled source of air pressure.

(b) Close the outlet end of valve with a suitable fitting.

(c) Pressurize the valve with the required pressure, and check for leakage for the specified time (see 4.3.3).

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4.4.4.7 Valve leakage.

(a) Connect the valve to a controlled source of air pressure.

(b) Close the outlet end of valve with a suitable fitting.

(c) Pressurize the valve with the required pressure and check for leakage for the specified time (see 4.3.3).

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The pressure control unit shall be cleaned and dried in accordance with MIL-P-116, using cleaning method C-1 and the most applicable of drying methods listed therein.

5.2 Packaging, level C. The pressure control unit shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.3 Packing, level C. The pressure control unit, packaged as specified in 5.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.5 Repair part. When this item is procured for storage and issue as a repair part preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. This item is intended for use in the M10-8 flame thrower for controlling air pressure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

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(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Caution. Due to the hazards inherent in testing with high pressures, suppliers are hereby notified that every precaution possible should be taken to prevent injury to personnel engaged in testing.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1040-A080

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